# SceneForge Outliner

User Guide · 1.0 beta for 3ds Max 2027 · by ForgePlugins

SceneForge Outliner is a fast scene browser for 3ds Max: one docked panel with a powerful filter, lenses, layers, isolation, reports and batch tools. This guide covers the beta build. Every command described here is in the panel; hover over any button for its tooltip.

> **Beta.** This build has no limits and needs no license key. The **License…** window shows the license as active for **Beta**. Please report anything odd: see Reporting bugs.

## Contents


1. Quick start
2. Installation
3. The panel at a glance
4. The object list
5. Selecting
6. The filter and its language
7. Lenses, row colours and explorer layouts
8. Layers
9. Groups, hierarchy and pivots
10. Isolation and viewport modes
11. Command palette, Menu button and undo
12. Scene reports
13. Scene health (Scene Doctor)
14. Scene rules
15. Batch rename and name by template
16. Object properties, materials, cameras and lights
17. Macros, states, views and configuration
18. Export and import
19. Performance and Light mode
20. Keyboard shortcuts
21. Preferences
22. Reporting bugs
23. Privacy & footprint
24. Known limitations in this beta

# 1. Quick start

Five minutes, seven steps. You need 3ds Max 2027 with SceneForge installed and any scene open (the bigger, the better).

### 1. Open the panel

- Right after installing, answer **Yes** to *Open the panel now to check that it works?*
- Later: main menu **SceneForge > SceneForge Outliner**. The **SceneForge** menu appears after you restart 3ds Max once.
- Or: **Customize > Customize User Interface**, category **SceneForge**, command **SceneForge Outliner**. Give it a keyboard key or put it on a toolbar.
- The panel opens as a docked window. The first opening loads the plugin and takes a moment; after that it just shows the panel.

### 2. Find an object with the filter

- Click the filter field at the top (its grey hint starts with *filter: name, type:Box ...*).
- Type part of a name, for example `chair`. The list shrinks while you type. Case does not matter.
- Try a key: `type:Box`, `layer:Walls*`, `-hidden:1` (the minus means NOT). A space between terms means AND.
- Or click a chip under the field: **Geometry**, **Lights**, **Cameras** ... **Hidden**, **Frozen**, **Selected**, **Last 10 min**. The **x** button clears the chips.
- Delete the text to see the whole scene again. The **?** button shows the full filter cheat sheet.

### 3. Select

- Click a row: the object is selected **in the scene**. `Ctrl`+click adds or removes, `Shift`+click selects a range.
- `Ctrl`+drag over rows selects every row you pass. (A drag without Ctrl moves objects: onto another row it links them as children.)
- `Ctrl`+`A` selects everything the list shows (the filter result, not the whole scene). `Ctrl`+`D` clears, `Ctrl`+`I` inverts.
- Select something in the viewport: the list highlights it and scrolls to it.

### 4. Hide and freeze

- The **State** column has an eye (visibility) and a snowflake (freeze). Click them.
- On a selected row the click acts on **all selected objects**; on any other row it acts on that row only.
- Or right-click a row: **Hide**, **Show**, **Freeze**, **Unfreeze**.
- `Ctrl`+`Z` undoes it in one step (also while the panel has focus).

### 5. The command palette: `Ctrl`+`P`

- Click inside the panel first. Outside the panel `Ctrl`+`P` is the 3ds Max Pan tool.
- Type what you want to do. Letters need not be next to each other: `brn` finds **Batch rename**.
- Arrows choose, `Enter` runs, `Esc` closes. Grey commands need something first (usually a selection).
- The **Menu** button (first on the top bar) has the same commands in groups, without typing.

### 6. Lenses: group the list

- On the top bar open the drop-down that shows **no grouping**.
- Choose **by layer (Layer Explorer)**, **by class**, **by material**, **by instance**, **by group** or **by face count**.
- A lens changes only the view, never the scene. The filter and the selection keep working.
- Under the layer lens, right-click a layer header for layer commands (hide, freeze, rename, ...).

### 7. Isolate

- Select the objects you want to work on, click into the panel and press `Ctrl` + `Alt` + `Q`.
- Everything else is hidden. A bar above the list shows **Exit isolation (Ctrl+Alt+Q)**.
- Press `Ctrl` + `Alt` + `Q` again (or click the bar) to bring back exactly what was visible before.
- Isolation is a mode, not an edit: `Ctrl` + `Z` does not leave it. The 3ds Max `Alt` + `Q` isolation is a separate tool.

> **Tip.** Every button and command has a tooltip. Hover for a second and the panel tells you what it does and what it will not do.

## 2. Installation

### Requirements

- **3ds Max 2027** on Windows. This package supports no other version: the installer checks the version and stops if it does not match.
- No internet connection. SceneForge works offline and sends nothing anywhere.
- About 3 MB of disk space in your 3ds Max user folder. No administrator rights are needed.

> **Beta.** This beta build has no limits and needs no license key. **SceneForge > License...** shows the license as active for **Beta**.

### Install

1. Save your work. Start 3ds Max 2027 (an empty or an open scene, both are fine).
2. Drag the SceneForge `.mzp` file you received from Windows Explorer onto a 3ds Max viewport.
   (Alternative: **Scripting > Run Script...** and pick the `.mzp` file.)
3. The installer checks the 3ds Max version, copies the files and checks every copied file.
4. It asks: *SceneForge Outliner has been installed. Open the panel now to check that it works?*
   **Yes** opens the panel right away (the installation summary then goes only to the MAXScript Listener, `F11`).
   **No** shows the summary window: the copied files, the path of this user guide, how to start the panel, where to report bugs and the line that uninstalls SceneForge.
5. **Restart 3ds Max** to get the **SceneForge** menu in the main menu bar. The commands in **Customize User Interface** work at once, without a restart.

Messages you may see instead:

| MESSAGE | MEANING |
|---|---|
| *This release of SceneForge is for: 3ds Max 2027. You are running it in 3ds Max ...* | Wrong 3ds Max version. Nothing was installed. |
| *The package is incomplete, NOTHING has been installed.* | The .mzp file is damaged or incomplete. Get the file again. |
| *SceneForge -- installation INCOMPLETE.* | A file could not be copied (for example it was locked). The window lists it under PROBLEMS. The previous files were left unchanged. Restart 3ds Max and install again. |

### The SceneForge menu

After a restart the main menu bar has a **SceneForge** menu with eight commands. The same commands are in **Customize > Customize User Interface**, category **SceneForge**, so you can give them keys, toolbar buttons or quad menu entries.

| COMMAND | WHAT IT DOES |
|---|---|
| SceneForge Outliner | Opens the panel (the first time it loads the plugin). |
| SceneForge - Scene reports | Opens the panel and its Reports window. |
| About SceneForge | Version, install folder, contact. |
| License... | The license window. In this beta it shows **Beta**. |
| Documentation | Opens this user guide in your web browser. It works offline. |
| Report a bug... | Copies your set-up details to the clipboard and opens the GitHub form or your e-mail. |
| Suggest a feature... | Opens the GitHub feature request form or your e-mail. |
| More plugins by the author | Opens forgeplugins.tech. |

### Where the files go

Everything goes into your own 3ds Max user folder, normally `%LOCALAPPDATA%\Autodesk\3dsMax\2027 - 64bit\ENU\` ( `ENU` is the language of 3ds Max). The summary window shows the exact paths.

| FOLDER | CONTENTS |
|---|---|
| `scripts\SceneForge\` | The plugin: panel, core, scene listener, start script, uninstaller, this user guide and `SceneForge_installed.txt` (the list of installed files). |
| `usermacros\SceneForge.mcr` | The eight commands of the SceneForge category. |
| `scripts\startup\` | A small script that adds the SceneForge menu when 3ds Max starts. |
| `plugcfg\SceneForge\` | Your settings, created on first use: `preferencje.json` (preferences), `skroty.json` (your shortcuts), `makra.json` (macros), saved views, scene rules, panel state and the panel log `dziennik_panelu.txt`. |

> **Language profiles.** 3ds Max keeps a separate user folder per language. If you install while running 3ds Max in English and later start it in another language, install SceneForge again in that language.

### Update to a newer build

- Drag the new `.mzp` onto a viewport, exactly like the first time.
- The installer replaces the plugin files and removes the older versions. Your settings, shortcuts, macros, saved views and rules in `plugcfg\SceneForge` are kept.
- If the panel was open, answer **Yes** to open the new panel, or simply **restart 3ds Max**. If you answer No, the previous panel keeps running until the next restart.

### Uninstall

1. **Scripting > Run Script...**, open the folder `scripts\SceneForge` (the installation summary shows its full path) and run the file `SceneForge_odinstaluj_v...ms`.
2. Or: **Scripting > New Script**, paste the lines below and press `Ctrl`+`E` (Tools > Evaluate All):

```
(
  SF_odinst = getFiles ((getDir #userScripts) + "\\SceneForge\\SceneForge_odinstaluj_v*.ms")
  sort SF_odinst
  if SF_odinst.count > 0 then fileIn SF_odinst[SF_odinst.count]
  else messageBox "SceneForge is not installed."
)
```

- **It removes:** the open panel (it first leaves isolation and the viewport filter mode and restores painted wireframe colours), the scene callbacks and the scene listener, the SceneForge columns in the native Scene Explorer, the loaded core, the files listed in `SceneForge_installed.txt`, the commands file, the startup menu script and the listener's own log file.
- **It keeps:** your settings in `plugcfg\SceneForge` and any file in the plugin folder that SceneForge did not put there (the folder then stays). The final window lists what was removed, what was kept and where; delete the kept files by hand if you do not need them.
- Data already saved inside your .max files stays in those files (see *Privacy & footprint*).
- The menu entry disappears after you restart 3ds Max.

### Troubleshooting

| PROBLEM | WHAT TO DO |
|---|---|
| There is no SceneForge menu | Restart 3ds Max once after installing. Until then use Customize User Interface, category SceneForge. |
| The panel does not open | Run **SceneForge Outliner** from Customize User Interface and look at the MAXScript Listener (`F11`) for a message. Then use **Report a bug...**. |
| `Ctrl`+`P` pans the viewport | Panel shortcuts work only while the panel has focus. Click a row or the panel first. |
| Home / End / Page Up / Page Down move the time slider | Click a row so the list has focus. A green label *Home/End/PgUp/PgDn → list* on the bottom bar shows when the list owns these keys. |
| The list does not follow the scene | Check the **auto** box on the top bar, or click **Refresh**. |
| The Faces column shows three dots | Not counted yet. Counting runs in small steps for the rows you see; scroll or wait. In **Light mode** automatic counting is paused. |
| A yellow bar says UNDO HAS NOT WORKED | Some script left an undo block open. When no operation is running, click **Close block**. |

| PROBLEM | WHAT TO DO |
| --- | --- |
| Settings were reset | An unreadable settings file is set aside as `<name>.damaged-<time>` and defaults are used. The status bar says so. |
| Uninstall left the folder | It holds files SceneForge did not install. They are listed in the final window. Delete the folder by hand if you do not need them. |

## 3. The panel at a glance

The panel is one docked window. From top to bottom:

| PART | WHAT IS THERE |
|---|---|
| Top bar | **Menu** button, filter field, **hist.** (filter history), **filter** box (switch the filter on or off without losing the text), **?** (cheat sheet), **Refresh**, **Classes**, **Layers**, **Sets** (put a class, layer or set into the filter), **Health**, **Reports**, the lens drop-down, the viewport drop-down (*view: no change / freeze the rest / hide the rest*), the row background drop-down, **Views** and the **auto** box (automatic refresh). |
| Chips | **All**, **None**, **Invert**, then the category chips (Geometry, Shapes, Lights, Cameras, Helpers, Space warps, Bones) and the state chips (Hidden, Frozen, Selected, Last 10 min), and **x** to clear them. |
| Info bars | Appear only when needed: isolation, viewport filter mode, painted wireframe colours, a stuck undo block, a parent hidden by the filter, pinned objects, the path bar and the view tabs. |
| Side bar | Icon buttons on the left: show or hide categories (right-click = only this category), show hidden, show frozen, layer lens, new layer, reports, rules, X-ray the rest, presentation mode, show only selected, select all, isolate, scene health, command palette, second view, preferences (right-click = keyboard shortcuts) and the cheat sheet. |
| Object list | The scene. Next to it the **layer list**; below it an optional **second view**. |
| Bottom bar | **Explorer:** ready-made layouts, **Set:** named selection sets, **Layer list** box, the **PgUp/PgDn:** step and a counter: rows shown / objects selected in the scene. |
| Status line | What the panel just did, and how long it took. |
| Performance row | The **Performance** button and the **Geometry:** status (see *Performance and Light mode*). |

Make the dock as narrow as you like: the bars wrap onto more rows. `Shift`+mouse wheel scrolls the list sideways. `Ctrl`+`Alt`+`X` (**Presentation mode**) hides everything except the filter field and the list and makes the font bigger; press it again to restore the layout.

## 4. The object list

### Columns

Right-click the column header and choose **Which COLUMNS to show...** (or the palette command **Which columns to show...**). Tick columns and move them with **Up** / **Down**. **Defaults** returns to the standard set.

| COLUMN | SHOWS |
|---|---|
| Name | The name, the hierarchy indent, the class icon and the expand triangle. Always on. |
| Class | The object class, not localised: Box, Editable_Poly ... |
| Layer | The object's layer. |
| State | Eye and snowflake. Click to hide or freeze (see *Selecting*). |
| Color | Wireframe colour as a swatch and a hex value. |
| Material | The material name. Empty = no material. |

| COLUMN | SHOWS |
|---|---|
| Faces | Face count at the top of the modifier stack. Counted lazily; three dots = not counted yet. |
| Instances | `x12` = twelve instances; `R3` = three objects on a common base object. |
| Mod. | Number of modifiers. Empty = none; three dots = not counted yet. |
| Depth | Depth in the hierarchy; 0 = scene root. |
| Children | Number of direct children. |
| Verts | Vertex count, read together with the faces. |
| Size | Largest bounding box dimension in world space, in scene units. |
| Flags | Letters: U hidden, Z frozen, S selected, G group head, g group member (handy in CSV). |
| Path | Ancestors from the root, separated by slashes. |
| Age | Time since the object last changed in this session: 12 s, 5 min, 2.3 h. |
| Tags | Labels from the `sf_tag` user property, with a coloured dot. |
| Note | Text from the `sf_note` user property. |

The default set is Name, Class, Layer, State, Color, Material, Faces and Instances. Columns that read geometry (Faces, Verts, Mod., Size) are filled only for the rows you can see, in small steps, so a big scene stays responsive.

### Sorting

Click a column header to sort, click again to reverse, a third time to go back to hierarchy order. Right-click the header for **Sort by**: name, class, layer, visibility, chronologically (creation order), face count, instance count, material, child count, depth in hierarchy, colour, hierarchy order. **Which criteria to show...** trims that menu; **Reverse current order** flips it.

### Hierarchy

- Click the triangle to expand or collapse a branch. Row menu **Hierarchy**: **Collapse all**, **Expand all**, **Expand to level N**, **Collapse / Expand all lens headers**. Collapsing works in hierarchy order only (no filter, no sorting, no lens).
- Guide lines show who is whose child (Preferences, **Hierarchy lines**).
- **Show selected object** (palette) expands the branches down to the selected object and scrolls to it.

### Drag and drop

| DRAG | RESULT |
|---|---|
| Rows onto another row | Links them as children of that object. Nothing moves in the viewport. |
| Rows onto an empty part of the list | Unlinks them (they become scene roots). |
| Rows onto a layer in the layer list | Moves them to that layer, in one undo step. |
| Rows onto a material header (by material lens) | Assigns that material. Onto **(no material)** removes the material. |

| DRAG | RESULT |
| --- | --- |
| A material header onto a row | Assigns the material to that row, or to the whole selection if the row is selected. |
| `Ctrl` + drag | Selects the rows you pass, with auto-scroll at the edges. Nothing moves. |

If you link objects to a parent the filter hides, a bar says so and offers **Show parent** (clears the filter and scrolls to it) or **Hide**.

### Renaming and double-click

- `F2` renames the current row in place. For many objects use `Ctrl`+`F2` (see *Batch rename*).
- Double-click a row: selects the object with its whole subtree (Preferences, **Double-click**). Double-click a camera: the active viewport looks through it.

### Second view, pins, path bar, view tabs

- **Second view: show or hide** adds a second list under the first one, with its own filter, lens and sorting. The selection is shared, because it belongs to the scene. **Second view: copy settings from the top one** starts it from the same point.
- **Pin (pin bar above the list)** in the row menu keeps objects at hand above the list, whatever the filter. Click a pin to select it; right-click to unpin.
- **Path bar: show or hide**: the ancestors of the current object as buttons; click one to select it.
- **View tabs: show or hide**: your saved views as tabs above the list.

## 5. Selecting

- A click on a row selects the object in the scene; `Ctrl`+click adds or removes; `Shift`+click selects a range. All of it is one undo step.
- `Ctrl`+drag selects the rows you pass over.
- A selection made in the viewport is highlighted in the list and the list scrolls to it (Preferences: **Selection from the scene** and **scrolling**).
- `Ctrl`+`A` **Select all in view** selects what the list shows after the filter. `Ctrl`+`D` **Clear selection** clears the whole scene. `Ctrl`+`I` **Invert selection** works within the view.
- Eye / snowflake in the **State** column: on a selected row it acts on every selected object; on an unselected row on that row only. One undo step.
- `Ctrl`+`Alt`+`S` **Show only selected**: the list shows only what is selected and follows every change. Your filter text stays.
- `Alt`+`Left` / `Alt`+`Right` **Previous selection** / **Next selection**: selection history like Back and Forward in a browser (up to 50 steps).
- Row menu **Select descendants**, **Select with descendants**, **Select similar to this object** (same class, faces and size within 20%, same material; the query lands in the filter so you can edit it).
- **Select by size...** (palette) asks for a condition such as `>1000` and selects the result.
- **Copy names** and **Paste names from clipboard and select**: names one per line (or separated by commas or semicolons) become a selection; missing names are listed in the status bar.

### Named selection sets

- Row menu **Selection > Create set from selection...**, **Add to set**, **Remove from set** (an emptied set is deleted).
- The **Set:** list on the bottom bar selects the objects of a set. The number in brackets is the member count; an exclamation mark means part of the set points at objects that no longer exist.
- Filter by set with `set:Name`. The panel warns once about a set without a name.

## 6. The filter and its language

The filter field accepts a plain piece of a name, and it also understands a small query language. The **?** button shows the same cheat sheet inside the panel. While you type, the field suggests keys, and after the colon the values found in this scene.

### Combining terms

| WRITE | MEANS | EXAMPLE |
|---|---|---|
| space | AND | `type:Box layer:Walls` |
| - at the start of a term | NOT | `-hidden:1` |
| ( a \| b ) | OR (brackets can be nested) | `(type:Box \| type:Sphere)` |

### Names

| WRITE | FINDS |
|---|---|
| `word` | Objects whose name contains the word (case-insensitive). |
| `"Box001"` | Exactly this whole name. |
| `/expression/` | A regular expression on the name. |
| `*` and `?` | Wildcards in names and in key values: `layer:Walls_*`. |
| `~word` | Fuzzy search: one typo, missing letter or extra letter still matches ( `~teapot` finds Teapto01). Words shorter than 4 letters are not fuzzed. |

### Keys

| KEY | MEANING | EXAMPLE |
|---|---|---|
| `type:` (also `class:` ) | Object class | `type:VRay*` |
| `layer:` | Layer | `layer:SF_Root_0*` |
| `mat:` | Material name | `mat:Chrome` |
| `set:` | Named selection set; `set:` alone = in any set | `set:Interior` |
| `cat:` | Category: Geometry, Shape, Light, Camera, Helper, SpaceWarp, System | `cat:Light` |
| `hidden:` `frozen:` `selected:` | 0 or 1 (empty = 1) | `hidden:0` |

| KEY | MEANING | EXAMPLE |
|---|---|---|
| `group:` | head, member, 1 or 0 | `group:head` |
| `poly:` | Face count after the modifier stack | `poly:>100k` |
| `children:` | Number of children | `children:>0` |
| `depth:` | Depth in the hierarchy | `depth:>2` |
| `instances:` | How many copies share the object | `instances:>1` |
| `size:` | Largest bounding box dimension (world, scene units) | `size:>1000` |
| `changed:` | Changed within a time window (s, m, h, d) | `changed:<10m` |
| `tag:` | Tag from `sf_tag` ; `tag:*` any, `tag:0` none | `tag:urgent` |
| `note:` | Word in the note `sf_note` | `note:check` |
| `map:` (also `tex:` ) | Material uses a map whose file matches | `map:*.hdr` |
| `is:` | A trait, see below | `is:missingmat` |

Numbers: `>100` `<10` `>=5` `<=5` `=0` `100..500` `100` , with `k` and `m` suffixes ( `poly:>100k` ). Put a value with spaces in quotes: `set:"Living room furniture"` .

### is: traits

| TRAIT | FINDS |
|---|---|
| `is:changed` | Added or changed since the baseline remembered in Reports > Snapshot comparison |
| `is:recent` | Changed in the last 10 minutes |
| `is:missingmat` | No material |
| `is:orphan` | No parent and no children |
| `is:nolayer` | No layer |
| `is:root` / `is:leaf` | Root / leaf of the hierarchy |
| `is:instance` / `is:reference` / `is:unique` | Shares the object / shares only the base object / occurs once |
| `is:selected` `is:hidden` `is:frozen` | Selected, hidden, frozen |
| `is:group` `is:grouphead` `is:groupmember` | Groups, group heads, group members |

> **Not there yet.** `is:zeroscale` , `is:nonuniform` , `is:faraway` and `has:...` keys are not supported in this beta. The filter says so with an error instead of showing an empty list. Use **Health** for scale and distance problems.

### Examples

```
type:VRayLight layer:02_* poly:>100k -hidden:1
(layer:A | layer:B) instances:>1
cat:Light -hidden:1
mat:Chrome set:Interior
~teapot
```

### Chips, history and switches

- Category chips combine with OR (an object has one category), state chips with AND: **Lights** + **Hidden** = hidden lights. Category chips show how many objects of that kind the scene has. **All** / **None** / **Invert** work like Display in the Scene Explorer; **x** clears the chips and keeps the text. Hide the chip bar in Preferences or with **Filter chips: show or hide**.
- **hist.** lists the last 20 queries the filter accepted; click one to use it again. The history is kept between sessions.
- The **filter** box switches filtering off without deleting your query.
- **Filter delay** (Preferences) sets how long the panel waits after the last key (250 ms by default).
- A wrong query shows an error message, never a silent empty list. An unknown set name lists the existing sets.

## 7. Lenses, row colours and explorer layouts

A lens regroups the list. It changes only the view; the scene is untouched.

| LENS | GROUPS THE LIST |
|---|---|
| no grouping | Plain parent and child hierarchy. |
| by layer (Layer Explorer) | By layer, including empty and nested layers. Right-click a layer header for the layer menu. |
| by class | One header per object class. |
| by material | One header per material, plus **(no material)**. |
| by material (tree) | Multi/Sub-Object materials opened into their sub-materials. |
| by instance | One header per shared object, its instances under it. |
| by group | One header per nearest group head; the rest under **(outside group)**. |
| by face count | Headers by order of magnitude: below 1 000, 1 000 to 9 999 ... The first use counts faces of every object. |

- Collapse or expand all headers with **Collapse lens headers** / **Expand lens headers**.
- **Lens at startup** (Preferences) chooses the lens applied every time the panel opens.
- The row background drop-down colours rows *by layer*, *by class*, *by material* or *by object color*, so group boundaries show without reading.
- The **Explorer:** list on the bottom bar has ready layouts that set filter, lens and colours at once: Whole scene, Layers (Layer Explorer), Lights (Light Explorer), Cameras, Geometry, Shapes and splines, Bones and animation, Particles and deformers, Helpers, Containers (Container Explorer). Below them are your own layouts, **Save current layout as...** and **Manage layouts...**.

## 8. Layers

### The layer list

- Shows every layer with its object count; the current layer is bold. Hide it with the **Layer list** box on the bottom bar or **Layer list: show or hide**.
- Click a layer: puts `layer:name` into the filter. Double-click: makes it current.
- Drag objects from the list onto a layer to move them. Drag a layer onto another layer to nest it; drop it on an empty part to move it back to the root. 3ds Max refuses loops (a parent under its own child).
- `Alt`+click a layer: **solo**. Only that layer stays on; `Alt`+click again restores the previous layout. One undo step.

### Layer menu (right-click a layer)

- **Hide layer** / **Show layer**, **Freeze layer** / **Unfreeze layer**, **Solo: only this layer (Alt+click)** / **End solo (restore layers)**.
- **Select the layer's objects** (also those the filter hides), **Set as current**, **Put into the filter as layer:**.
- **New layer...**, **New layer UNDER ...**, **Layer color...**, **Rename layer...**.
- **Delete layer** (3ds Max refuses a layer that holds objects) and **Delete layer with sublayers...** (asks what to do with the contents: move to layer 0, delete with the layers, or touch only empty layers; one undo step brings everything back).
- **Face budget for layer...**: the layer list shows faces / budget and turns red above it. 0 removes it.

### Layers from the row menu

Right-click objects, **Layer**: **To the current layer**, **Choose layer...** (a window with a name filter, quick with many layers), **New layer from selection...**, **New empty layer...**, **Set the selection's layer as current**, or pick a layer from the list.

### Layer states `Ctrl`+`Shift`+`L`

Saves the visibility, freeze, lock and colour of all layers, and the current layer, under a name. **Restore checked parts** restores only what you tick; **What changed?** lists the differences layer by layer. The states are saved in the scene file. The same window also lists 3ds Max **Scene States** (**Restore Scene State**, **Save Scene State (layers)...**).

## 9. Groups, hierarchy and pivots

### Groups

Row menu **Group**: **Group...** (asks for a name), **Ungroup** (one level), **Explode (with nested)**, **Open group**, **Close group**, **Detach from group**. The palette also has **Group selected** and **Ungroup**. Filter with `group:head`, `group:member` or the **by group** lens.

### Hierarchy

**Unlink from parent** makes the selected objects scene roots without moving them. Linking is done by dragging rows onto a row.

### Pivot

Row menu **Pivot** (also in the palette as **Pivot: ...**). The objects do not move, only their pivot.

| COMMAND | PIVOT GOES TO |
|---|---|
| Center to object | The centre of the bounding box (like Hierarchy > Pivot > Center to Object). |
| To bottom (on the floor) | The middle of the lowest face of the bounding box in world space. |
| To top | The middle of the top of the bounding box. |
| To bounding-box corner | The minimum corner; handy for tiling and arrays. |
| To world origin | [0,0,0]. |
| To parent's pivot | The parent's pivot position; objects without a parent are skipped and counted. |
| To common centre of selection | One shared pivot: the centre of the whole selection. |
| Align to object / Align to world | Rotates the pivot axes. |
| Copy orientation from first selected | The pivot orientation of the first selected object, given to the others. |
| Reset pivot / Reset transform / Reset scale | The 3ds Max reset commands. |

> **Note.** Changing pivot orientation on an object with non-uniform scale also moves and deforms it. 3ds Max does this without warning; SceneForge counts such objects and asks first. Run **Reset scale** before, if needed.

## 10. Isolation and viewport modes

| COMMAND | WHAT IT DOES |
|---|---|
| Isolate selected `Ctrl`+`Alt`+`Q` | Hides everything except the selection and remembers the previous visibility of objects and layers. Press again to exit. |
| Isolate with descendants | The same, with the whole subtrees of the selected objects. |
| Exit isolation | Restores exactly what was visible before, including objects you had hidden by hand and layers that were off. |
| view: freeze the rest `Ctrl`+`Alt`+`F` | Freezes everything the filter does not show. Those objects stay visible as a reference but cannot be clicked. |
| view: hide the rest | Hides everything the filter does not show: isolation driven by the filter. |
| Release view `Ctrl`+`Alt`+`F` | Leaves the viewport mode and restores the previous state. |
| X-ray the rest | Selected objects stay solid, everything else becomes See-Through. Click again to undo it on exactly those objects. |

- Isolation and the viewport filter modes are **modes, not edits**: `Ctrl`+`Z` does not leave them. Use the bar above the list or the same shortcut.
- While the filter drives the viewport (hide the rest), isolation is not available; release the view first.

### Viewport colouring

**Viewport: color objects by layer / by class / by material / by face count** (palette) replaces the wireframe colour of every object. By face count gives a green to red heat map of the scene's weight. This **is a scene change**: a bar above the list reminds you, and **Restore wireframe colors** (bar button or palette) brings back the originals in one undo step. If you save the file while painted, the painted colours are saved.

## 11. Command palette, Menu button and undo

`Ctrl` + `P` (while the panel has focus) opens the **command palette**: every panel command with a short description. Type a few letters in any order of words; letters need not be adjacent. While the field is empty or holds one or two letters, the commands you used last come first. A grey command shows why it is not available now.

The **Menu** button holds the same commands in groups: **List and view**, **Viewport**, **Layers**, **Selection**, **Material**, **Objects**, **Reports and rules**, **Settings and tools**, **Configuration**, **Help** and **Other** (for example the **Pivot** commands). Shortcuts are shown on the right.

The **Help** group has **About SceneForge**, **Documentation**, **Report a bug...**, **Suggest a feature...** and **More plugins by the author**, the same as the SceneForge menu.

### Undo and redo

Every panel operation that changes the scene is one entry in the 3ds Max undo history. While the list has focus `Ctrl` + `Z` undoes and `Ctrl` + `Y` or `Ctrl` + `Shift` + `Z` redoes. In the filter field and the name editor `Ctrl` + `Z` undoes typing. **Undo last operation** and **Redo undone operation** are also in the palette.

If an undo block stays open (for example after an interrupted script), 3ds Max silently stops recording undo. The panel notices it and shows a yellow bar with **Close block**. The palette command is **Close the stuck undo block**. Use it only when no operation is running.

## 12. Scene reports

**Reports** button, `Ctrl` + `Shift` + `R`, or **SceneForge > SceneForge - Scene reports**. Nothing is computed when the window opens: each tab has its own button, so opening the window never stalls 3ds Max. Most tabs can select what they show and save it as CSV.

| TAB | WHAT IT ANSWERS |
|---|---|
| Geometry duplicates | **Find duplicates** lists groups of objects with identical geometry. **Check geometry** compares vertex positions; **Merge into instances** turns a group into instances, **Merge ALL** does it for every group (groups whose geometry differs are skipped and named). **Select group**, **CSV export**. |
| Scene weight | **Compute scene weight**: faces and objects per layer and per class; **Select the chosen**. |
| Textures | **Check textures**: every bitmap file, its size on disk and whether it is missing. **Relink missing...** searches a folder (with subfolders) for files of the same name. **Select objects using it**. |
| Naming | Rules as `name\|regular expression`, one per line. **Check names**, **Select violators**, **Save rules**, **Preview** and **Fix names**. Before fixing, the old names are written to a CSV file next to the scene. |
| Snapshot comparison | **Remember baseline**, later **Compare**: added, removed and changed objects. **Show changed in the list** puts `is:changed` into the filter. |

| TAB | WHAT IT ANSWERS |
|---|---|
| Timeline | Recent scene events with time and name (added, deleted, renamed, hidden, undo and redo). **in the list** puts `changed:<N m` into the filter. |
| Scene card | **Count**, then **Save HTML...**: one HTML page with the scene's numbers (objects, layers, faces, heaviest layers and classes, health, textures, duplicates) for someone without 3ds Max. |

## 13. Scene health (Scene Doctor)

The **Health** button (also **Scene health** in the palette and the row menu) scans the whole scene for things that raise no error but spoil the work. Findings are listed with counts, most serious first; zero findings stay on the list in grey, so *checked and clean* is visible.

- **Show these objects** (or double-click a finding) selects them and puts `is:selected` into the filter.
- **Fix** repairs a finding when the fix is unambiguous. A **Scene Doctor** confirmation says how many objects it touches. One undo step reverts it.
- **Recompute** rebuilds the snapshot and checks again.

| FINDING | FIX |
|---|---|
| Zero scale | Scale 1,1,1 (only objects whose own scale is zero). |
| Negative scale | Scale 1,1,1 (mirroring disappears; check normals). |
| Far from the origin | Moves the objects together so the centre of their pivots lands at 0,0,0. |
| No material | Renderable geometry gets a shared Physical Material named `SceneForge_Default`. |
| Duplicate names | The first keeps its name, the next ones get a free number. |
| Hidden and frozen at once | Unhide and unfreeze. |
| Empty layers | Deletes layers without objects and sublayers (never the current one or layer 0). |
| Meshes without faces | Deletes them (particle systems, procedural objects and groups are kept). |
| Non-uniform scale | Reported only; no automatic fix. |

## 14. Scene rules

**Scene rules...** (palette, side bar) stores rules made of a filter query and an action. A rule works on the **whole scene**, also on objects the list does not show.

- Fields: **Name**, **Query** (filter language), **Action**, **Argument** (the layer name for Move to layer).
- Actions: Hide, Show, Freeze, Unfreeze, Select in scene, Display as box: on / off, See-Through: on / off, Disable rendering, Enable rendering, Move to layer (argument).
- **Test** counts matching objects and changes nothing. **Run** runs one rule; **Run all** runs them top to bottom after showing the counts. Each run is one undo step.
- **Automatic rules** (Preferences, off by default): after opening a file and after an import the panel waits 1.5 s and runs all rules (after an import only on the new objects).

## 15. Batch rename and name by template

### Batch rename `Ctrl` + `F2`

Works on the selected objects. The steps always run in this order: base name, trim, find and replace, letter case, number, prefix and suffix.

- **Base name** (empty keeps the names), **Prefix**, **Suffix**, **Trim characters** *from the front* / *from the end*.
- **Find** / **Replace with**, with **regular expression** (groups `$1`, `$2` work) and **case sensitive**.
- **Letter case**: no change, UPPERCASE, lowercase, Each Word Capitalized.
- **Numbering**: start, step, digits. The order is the row order in the list, so sort the list first.
- The preview shows *now* and *after change*, counts repeated names and skips rows that would get an empty name. **Rename** applies it to all selected objects in one undo step and saves the old names to a CSV file next to the scene.

### Name by template

Row menu **Name by template...**: a template with tokens `{name}`, `{class}`, `{layer}`, `{mat}` and a counter `{##}` (as many digits as hashes), for example `{layer}_{class}_{##}`. Preview, then **Rename**, one undo step.

## 16. Object properties, materials, cameras and lights

### Object properties `Alt` + `Enter`

The General properties of 3ds Max for many objects at once: Hidden, Frozen, See-Through, As box, Backface cull, All edges, Vertex ticks, Trajectory, Ignore in Zoom Extents, Frozen in material color, vertex colour display, Renderable and more. A field whose value differs between objects shows a mixed state and is not written until you change it.

### Display and rendering (row menu)

- **Hide**, **Show**, **Freeze**, **Unfreeze**, **Renderable: on** / **off**.
- **Display**: As box, See-through, Backface cull, All edges, Vertex ticks, Trajectory, Frozen in color (on / off).
- **Inherit from layer**: display, motion blur and colour, each *take from layer* or *set on the object*. An object that takes a property from its layer ignores its own setting.
- **Object color...**, **Zoom in viewport**.

### Materials

- Row menu **Material**: **Open material in editor** (Slate or classic, whichever mode is on), **Put material into the active slot of the classic editor**, **Assign material from slot** (the 24 classic slots), **Assign material from Slate editor**, **Copy material from the first selected to the rest** (the same material, not a copy), **Replace this material in the whole scene with...**, **Remove material**, **Show objects with this material (filter)**.
- Under the **by material** lens, drag rows onto a material header to assign it.

### Instances

Row menu **Instances**: **Select all instances of this object**, **Select instances TOGETHER with references**, **Show only instances (filter)**, **Count instances in the scene**.

### Cameras and lights

- **Create camera from view** (palette): a camera that matches the active perspective view, in the class of the current renderer (V-Ray Physical Camera, Corona Camera, FStorm Camera, or a Physical Camera for Arnold and the others). The viewport switches to it. One undo step.
- **Look through this camera** (or double-click a camera row).
- **Turn lights on** / **Turn lights off** for the selected lights, one undo step.
- **Render selected cameras in sequence...** (EXPERIMENTAL, enable in Preferences, **Render queue**): each selected camera is rendered with the current renderer to `folder\camera_name.png`. 3ds Max is blocked until the queue ends. Save the scene first.

### Tags and notes

Row menu **Tags...** (labels separated by commas) and **Note...**. They are stored as user properties `sf_tag` and `sf_note` (visible in Object Properties > User Defined) and saved with the .max file. Columns **Tags** and **Note**, filter `tag:` and `note:`. Empty text removes them.

### Containers

**Containers in the scene...** lists containers with their state and source file. The row menu **Container** has Open, Close, Load contents, Unload contents, Reload, Update from source file, Make unique and Merge source into scene. Commands that need a source file refuse (and say why) when the container has never been saved.

### Columns in the native Scene Explorer

**Max explorer with SceneForge columns** opens a 3ds Max Scene Explorer with extra columns (instances, children, depth, subtree size and a short note). **Add SceneForge columns to your own explorer** appends them to the explorer you use; **Remove SceneForge columns from explorers** takes them off.

### Delete

Row menu **Delete** deletes the selected objects in one undo step and asks first when there are many.

## 17. Macros, states, views and configuration

### Macros

- **Macro: start recording**, then work: filters, palette, Menu and shortcut commands and row menu actions are noted. **Macro: stop and save...** asks for a name.
- Each macro becomes a palette command **Macro: name**, so it can also get its own shortcut.
- **Macros: list, run, delete...** shows the steps, runs or deletes macros (stored in `makra.json`).

### Object visibility states

**Object visibility states...**: named snapshots of which objects are hidden and frozen. **Save current state...**, **Restore** (exactly that layout, one undo step), **Delete state**. Saved in the scene file.

### Saved views

- A view remembers filter, sorting, lens, column widths and the panel split. It does not remember the selection.

- **Views** button or **Save current view...**; the first nine load with `Ctrl`+`1` to `Ctrl`+`9`. **Manage views...** renames, reorders, overwrites and deletes them.
- Views are stored in the plugin settings, not in the scene file.

### Configuration

- **Configuration: remember panel state on/off**: filter, sorting, lens, column widths, split and the second view are saved by themselves and come back when the panel opens. On by default.
- **Configuration: save to file...** / **load from file...**: preferences, layout and views in one JSON file, to keep several set-ups or move them to another computer.
- **Configuration: save current as default** / **load default**.
- **Panel settings in the scene file: save now** writes filter, lens, columns, pins and budgets into the scene (see *Privacy & footprint*).

## 18. Export and import

| COMMAND | WHAT IT DOES |
|---|---|
| Export list to CSV... | Saves what the list shows (visible columns, current filter and order), or only the selected rows. Separator: semicolon, comma or tab; optional header row; preview before saving. |
| Export layers to files... | Every checked layer with objects into its own `.max` (saveNodes) or `.fbx` (current FBX settings) file, named after the layer. |
| Annotations: export to JSON... / import from JSON... | Tags, notes and pins of all objects, matched by name. Missing names are listed. The import is one undo step. |
| Merge file with name check... | Reads object names from a .max file without opening it, lists name collisions with the scene and merges with a prefix for the colliding ones. The merge cannot be undone with Ctrl+Z; the merged objects stay selected. |
| Differences between two .max files... | Object names in file A and file B (or the current scene) without opening them: only in A, only in B, common. **Save CSV...**. A renamed object shows as removed plus added. |

## 19. Performance and Light mode

The **Performance** button sits under the status line. Next to it, **Geometry:** tells you the state of the automatic face and vertex counting (for example *not checked*, *pending*, *automatic counting paused*).

| MENU ITEM | WHAT IT DOES |
|---|---|
| Light mode | A switch. Pauses the automatic geometry counting: the Faces, Verts, Size and Mod. columns, the layer face budgets and the footer counters. Use it on very heavy scenes. The setting is remembered. |
| Recalculate selected geometry | Counts faces for the selected objects now, on demand. |
| Measure slow objects... | Measures how long 3ds Max takes to read the geometry of each object, for the *Selected objects* or the *Whole scene*. Slowest first. **Measure**, **Cancel**, **Select object**. A scene change cancels the measurement. |

| MENU ITEM | WHAT IT DOES |
| --- | --- |
| Record callback diagnostics | A switch. Writes timings of the panel's internal events to the panel log. Turn it on only when you report a slowdown or a freeze, and off afterwards. |

- **Footer counters** (Preferences, EXPERIMENTAL, off by default): objects, faces, triangles and vertices for the selection and for the whole scene. The scene total must evaluate every object once, which can take seconds on a big scene.
- On long operations the panel shows a wait cursor and a message in the status line.
- A hidden panel does no geometry work. The list draws only the rows you can see.

## 20. Keyboard shortcuts

Panel shortcuts work while the panel has focus (click a row or the panel). Outside the panel 3ds Max keeps all its own shortcuts.

| KEY | DOES |
|---|---|
| `Ctrl`+`P` | Command palette |
| `F2` | Rename in place |
| `Ctrl`+`F2` | Batch rename |
| `Alt`+`Enter` | Object properties |
| `Ctrl`+`A` / `Ctrl`+`D` / `Ctrl`+`I` | Select all in view / clear selection / invert |
| `Alt`+`Left` / `Alt`+`Right` | Previous / next selection |
| `Ctrl`+`Alt`+`Q` | Isolate selected / exit isolation |
| `Ctrl`+`Alt`+`F` | Filter drives the view: freeze the rest / release view |
| `Ctrl`+`Alt`+`S` | Show only selected |
| `Ctrl`+`Alt`+`X` | Presentation mode |
| `Ctrl`+`Shift`+`R` | Scene reports |
| `Ctrl`+`Shift`+`L` | Layer states |
| `Ctrl`+`1` … `Ctrl`+`9` | Load saved view 1 to 9 |
| `Ctrl`+`Z` / `Ctrl`+`Y` / `Ctrl`+`Shift`+`Z` | Undo / redo (list has focus) |
| `Home` `End` `PgUp` `PgDn` | Scroll the list (list has focus); the step is set with **PgUp/PgDn:** on the bottom bar |
| `Shift` + wheel | Scroll the list sideways |
| `Alt` + click a layer | Layer solo |
| `Ctrl` + drag | Select rows by dragging |

### Your own shortcuts

**Custom keyboard shortcuts...** (palette, Preferences > **Keyboard shortcuts...**, or right-click the Preferences icon on the side bar). Every panel command is listed, including your macros and the Pivot commands. Double-click a command (or **Assign shortcut to selected...**) and press the keys; `Delete` in the table clears it. A shortcut needs `Ctrl` or `Alt` or must be a function key. It works while the panel has focus and, there, takes priority over a 3ds Max shortcut with the same keys.

To open the panel itself with a key, use 3ds Max: **Customize User Interface**, category **SceneForge**, command **SceneForge Outliner**.

## 21. Preferences

**Preferences...** (palette, Menu, side bar icon). The settings are saved in `plugcfg\SceneForge\preferencje.json`.

| SETTING | MEANING |
|---|---|
| Refresh | Refresh the list automatically after scene changes (the same as the **auto** box). |
| Filter delay | Wait after the last key before filtering (default 250 ms). |
| Undo guard | Warn about a stuck undo block; samples before the alarm; optionally close a stuck block by itself after a quiet period. |
| Row height | List row height (0 = system default). |
| Lens at startup / Row background at startup | Applied every time the panel opens, or as in the last session. |
| Safety | Ask before deleting layers with sublayers. |
| Double-click | Double-click selects the object with its whole subtree. |
| Selection from the scene / scrolling | Highlight in the list what is selected in the scene, and scroll to it. |
| Scene file | Save panel settings in the scene file (on by default). |
| Footer counters | EXPERIMENTAL face counters for the selection and the scene. |
| Render queue | EXPERIMENTAL: enables **Render selected cameras in sequence**. |
| Expert mode | EXPERIMENTAL: keep the panel visible in 3ds Max Expert mode (`Ctrl` + `X`). |
| Hierarchy lines | Draw hierarchy guide lines. |
| Filter chips | Show the chip bar under the filter field. |
| Automatic rules | Run all scene rules after opening a file and after importing (off by default). |
| Keyboard shortcuts... | Opens the custom shortcuts window. |

## 22. Reporting bugs

Thank you for testing. Every report is read.

### How to report

- In 3ds Max choose **SceneForge > Report a bug...** (also in the panel: Menu > Help, or the palette).
- It copies your set-up details to the clipboard and asks: **Yes** opens the GitHub bug form, **No** opens your e-mail program. Nothing is sent automatically.
- Describe what you did, what you expected and what happened instead. Then paste the details (`Ctrl` + `V`). A screenshot helps a lot.
- GitHub: github.com/Forge-Plugins/SceneForge/issues  E-mail: forgeplugins@gmail.com
- Ideas: **SceneForge > Suggest a feature...**, the same two ways.

> **Check before you post publicly.** The copied details contain the SceneForge and 3ds Max versions, the 3ds Max language, the Windows version, the install folder (it includes your Windows user name), the number of objects in the scene and the last lines of the panel log, which can contain file paths and object names. Read the text before you paste it on GitHub. For client scenes, send the report by e-mail instead.

### What helps

- The steps to repeat the problem, and how often it happens.
- Roughly how many objects and layers the scene has, and which renderer is active.
- For a freeze or a slow operation: switch on **Performance > Record callback diagnostics**, repeat the problem, switch it off, then send the panel log `dziennik_panelu.txt` from `plugcfg\SceneForge`.
- Any message from the MAXScript Listener (`F11`).

## 23. Privacy & footprint

### Network

SceneForge works offline. It does not connect to the internet, sends no data and has no telemetry. A browser or e-mail program opens only when you choose **Report a bug...**, **Suggest a feature...** or **More plugins by the author**.

### What can end up in your .max files

| WHAT | WHEN | HOW TO AVOID OR REMOVE IT |
|---|---|---|
| Panel settings: filter, lens, visible columns, pins, face budgets (in the scene root's appData) | Every time you save, while **Scene file** is on (default). Also with **Panel settings in the scene file: save now**. | Preferences > **Scene file** off. |
| Object visibility states | Only when you save one. | Delete the state in its window. |
| Layer states | Only when you save one. | Delete the state in its window. |

| WHAT | WHEN | HOW TO AVOID OR REMOVE IT |
|---|---|---|
| A marker that automatic rules already ran on this file | Only with **Automatic rules** on. | Keep Automatic rules off (default). |
| Tags and notes: user properties `sf_tag`, `sf_note` | Only when you set them. | Set empty text, or edit Object Properties > User Defined. |
| Painted wireframe colours | Only after Viewport colouring, until you restore them. | **Restore wireframe colors** before saving. |
| Material `SceneForge_Default` | Only when you use the Health fix for missing materials. | Delete or replace it like any material. |

Everything else the panel does (isolation, viewport filter modes, X-ray, lenses, filters) lives only in memory and is not written to the file. Normal scene edits you make through the panel (renames, layers, hiding, materials ...) are of course saved like any other edit.

### Files on your disk

- Plugin files in `scripts\SceneForge`, `usermacros` and `scripts\startup` (see *Where the files go*).
- Settings, macros, views, rules and the panel log in `plugcfg\SceneForge`. The log is capped at 1 MB; the previous part is kept once.
- `scripts\SceneForge_events_log.txt`: written by the scene listener only if something goes wrong.
- CSV files you save, the old-names CSV next to the scene after renames, exported layers, rendered images and scene cards, only where you choose (or next to the scene).

## 24. Known limitations in this beta

- **3ds Max 2027 only.** This package does not install in other versions.
- **Very large scenes** (around 50 000 objects): selecting or deselecting everything can take a moment, because 3ds Max redraws the viewport.
- **Undo and redo after deleting very many objects** can take a long time. This is how 3ds Max itself handles that undo step.
- **Merge file with name check** cannot be undone with `Ctrl`+`Z` (3ds Max records no history for a merge). The merged objects stay selected, so you can delete them.
- Isolation and the viewport filter modes are modes, not undo steps.
- Panel shortcuts, including `Ctrl`+`P`, work only while the panel has focus.
- Features marked EXPERIMENTAL (footer counters, render queue, Expert mode) are not fully tested.
- The filter does not yet support `is:zeroscale`, `is:nonuniform`, `is:faraway` or `has:` keys.
- The user interface is in English only.